

THE STARSHIP GALILEO
DEEP SPACE TRAVEL AND COLONIZATION
INSTRUCTIONS MANUAL
THE GEMINI PROTOCOLS
PHASE 175
THE MAGNETIC SPRING
FOUNDRY-FREE REPULSION STACK

Like poles on a pole
soft at the top, hard at the bottom



the foundry stays home

RING STACK · VOLUME KNOB · DAMPING CLAUSE

A SPRING FROM MANIFEST STOCK

Mechanical system — v1.0 in force

GP-IM-175

Revision v1.0 — 23 August 2026

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VETTED BATCH 20 — PASS

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LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — INSTRUCTIONS MANUAL — PHASE 175

PHASE 175: THE MAGNETIC SPRING — FOUNDRY-FREE REPULSION-STACK SUSPENSION — STATUS: CURRENT — CLEAN PASS (VET BATCH 20). The kids' toy explains it all: ring magnets slid over a vertical pole, like poles facing, each ring floating on the repulsion of the one below — and on Earth the spacing is unequal, because gravity compresses the stack, which is itself a physics lesson for colony kids. Phase 175 makes the toy a suspension: a coil spring is a foundry product — alloy metallurgy, wire drawing, hot winding, tempering — a workshop tier the ship does not carry, while neodymium ring magnets are standard manifest stock by the crate. The repulsion stack is a progressive-rate spring (soft at the top of travel, firm at the bottom), foundry-free, rebuildable from stock. The colony path is the electromagnet variant: copper wire and iron cores from common stock, coil current as the volume knob — soft ride unloaded, firm ride hauling regolith, trimmed from the driver's seat — with the honest cost recorded: permanent magnets hold repulsion for free, electromagnets hold it for watts. Applications: rover corners, equipment isolation, contactless bearings that cannot rust because nothing touches, and the schoolroom kit. The vet passed it clean; the audit's damping clause rides in §2.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Manual GP-IM-175, v1.0 — 23 August 2026. The one hundred and seventy-sixth manual of the program — 176 of 290.

§0 — READ THIS FIRST (HOW TO USE THIS MANUAL)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the spring law as seated (contents line, the bodies — the toy, the foundry problem, the volume knob, the applications — the audit and provenance, verbatim) — §2 why repulsion is a spring (graded) — §3 the system on the pole — §4 the doctrine record — §5 the vet record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 175: **The Magnetic Spring — Foundry-Free Repulsion-Stack Suspension**. Stacks of like-poled neodymium ring magnets on a guide pole act as progressive-rate springs (soft at the top of travel, hardening with compression), replacing foundry-made coil springs the ship and colonies cannot manufacture. Equalized spacing in zero-G, silicone impact end-buffers, eddy-current damping from a copper guide pole, electromagnetic trim stages for dial-adjustable spring rate, and contactless bearings for instrument masts and weathervanes that cannot rust.'

THE CORE OBJECTIVE (verbatim): 'Core Objective: Elimination of Foundry-Made Coil Springs from Ship, Rover & Colony Suspension and Bearing Duty [*WORDING KILLED 15 AUGUST 2026 — author's order 'kill anything that says absolute or 100 percent'; original: '100%'*]

Vehicle & Equipment Baseline: Contactless Progressive-Rate Suspension Without Foundries

Live instruction, 30 July 2026. Third invention of the sitting — chosen by Kimi from Archer's three-offer menu (birthday cake / Hansel and Gretel / magnetic spring) at the close of the tether-doctrine discussion; the other two remain in his queue. Physics audit appended per file doctrine.

THE KIDS' TOY THAT EXPLAINS IT ALL (VERBATIM)

- **The device:** ring magnets slid over a vertical pole, like poles facing, so each ring floats on the repulsion of the one below. On Earth the spacing is unequal — gravity's weight compresses the lower gaps more, because the bottom magnet carries the whole stack. In space there is no weight; the gaps equalize and the stack works as a pure spring, with silicone or like impact buffers taking the end stops.
- **The cone variant:** taper the stack like the children's toy — reducing magnet size up the pole, the higher-power, lower-field, lighter rings on top. The light top end moves easily, exactly like the first compression stage of a progressive-rate spring, firming as travel deepens. What Earth suspension shops achieve with dual-rate wound coils and rubber helpers, the pole achieves with geometry.
- **The schoolroom reading:** on Earth it is a physics lesson — same stack, same magnets, different spacing under gravity than in free fall. For colony kids it is the first device that shows a force field doing mechanical work with nothing touching. Their tricycle rode a fan; their suspension floats on a field.

THE FOUNDRY PROBLEM — WHY THE COIL STAYS HOME (VERBATIM)

- **A coil spring is a foundry product.** Spring steel needs alloy metallurgy, wire drawing, hot winding, and tempering — a workshop tier the ship does not carry and a young colony will not reach for years. Every coil spring on a rover corner is an Earth import with a fatigue life and a shipping manifest.

- **The fleet already carries the alternative.** Neodymium ring magnets are standard manifest stock — the Mayernick arrays run on them by the crate. A magnetic spring is assembled, not forged: pole, rings, buffers, done. Carbon/ferrite magnets do not hold the field strength for this duty; neodymium does — Archer's call, and the audit agrees.
- **Colony path:** when local manufacture begins, the electromagnet variant is the colony-sustainable route — copper wire and iron cores are makeable from common stock long before a rare-earth sintering line exists. The permanent-magnet stacks run from manifest; the wound stacks grow with the colony.

THE ELECTROMAGNET VOLUME KNOB (VERBATIM)

- **Turn it on with the vehicle.** Electromagnet repulsion stages energize with the ignition — no leftover kits, no separate infrastructure. Park, and the vehicle settles onto its silicone buffers; start, and the suspension wakes up.
- **Turn up the volume.** Coil current sets the spring rate — soft ride unloaded, firm ride hauling regolith, trimmed from the driver's seat or the ride computer. A spring whose rate is a dial is a suspension shop in a knob.
- **The honest cost, recorded:** permanent magnets hold repulsion for free, forever; electromagnets hold it for watts, continuously. The hybrid is the spec — permanent-magnet stack as the baseline spring, electromagnetic trim as the rate adjustment — so a dead electrical system degrades the ride, never the vehicle.

APPLICATIONS LEDGER (VERBATIM)

- **Rover and vehicle corners (ship & colony):** repulsion-stack corners with silicone end buffers — foundry-free, rebuildable from manifest stock, progressive rate by stack geometry. Archer's own caveat honored: too lightweight for full Earth vehicle duty — on Earth, steel wins on mass-per-force; on a low-mass colony rover or a ship's utility cart, the stack is in range and the foundry is absent.
- **Equipment isolation:** instruments, genesis pods, galley machinery, and the Phase 168 steam-cook line ride on small magnetic stacks — contactless isolation with no elastomer to outgas and no metal fatigue to log.
- **Contactless bearings — the weathervane answer:** a levitated pivot that cannot rust because nothing touches. Archer's instinct is industrial fact: magnetic bearings already run in compressors and flywheels on Earth; the colony weathervane, anemometer, and sensor masts get them by default.
- **The schoolroom kit:** the cone stack, boxed, as the colonies' first home-built physics demonstration — gravity on, gravity off, spacing tells the story.

PHYSICS NOTE — THE AUDIT (KIMI: AFFIRMED, WITH THE DAMPING CLAUSE) (VERBATIM)

The repulsion stack is a genuine spring with a genuinely useful law: coaxial magnet repulsion rises steeply as the gap closes (dipole-dipole far-field runs roughly as the inverse fourth power of separation), which is precisely a progressive, hardening spring rate — soft at the top of the travel, firm at the bottom. Archer identified this from the toy itself ('the top end moves easily... getting harder and harder'); the toy and the physics agree. His Earth-versus-space observation is exact: under gravity the stack's gaps compress downward in proportion to carried weight, and in free fall they equalize — the stack is a working gravimeter, which is why it belongs in the schoolroom. Two completions the audit adds, per doctrine. First, damping: a spring without a damper is an oscillator, and magnets alone provide almost none — but the guide pole answers it. Make the pole a copper or aluminium tube and every stroke of the rings induces eddy currents in the wall, producing velocity-proportional drag with zero contact: the same physics as a roller-coaster's magnetic brake fin. Pole as guide, pole as damper, silicone as end-stop — the suspension is complete. Second, stability: coaxial repulsion stacks are laterally unstable (Earnshaw's theorem forbids stable levitation by static permanent magnets alone), and the pole is not optional — it is the mechanical constraint that makes the system lawful. Bore friction at the rings is the only wear point; PTFE or silicone bore liners close it. Mass honesty: per unit of force, wound steel still beats magnets on Earth, which is why this phase targets the places the foundry cannot reach — ship, colony, and every bearing that should never rust.

- **AMENDMENT — VETTING BATCH 174-184 (THIRD-PARTY PASS, SEATED 10 AUGUST 2026): THE EARNSHAW VIGILANCE, MADE EXPLICIT.** The device bullet's phrase 'works as a pure spring' in zero-G must never be read as lateral self-stability: static permanent magnets alone CANNOT provide stable lateral levitation — the pole is therefore NOT decorative. It supplies the mechanical constraint that makes the machine sound (and the conductive pole doubles as the eddy-current damper). The original wording stands untouched per doctrine; this vigilance rides beside it so no reader ever upgrades the spring into a levitation claim. Credit: the third-party vetting pass, which graded the phase PASS and asked that this distinction be preserved in so many words.

ORIGIN OF THOUGHT — PHASE 175 (VERBATIM)

Archer, 30 July 2026: "ok on earth gravity can be the enemy of physics, ring magnets on a pole stacked, dont keep equal space the weight is still there, but not in space, so spring coils need foundries we wont have on the ship nor on new colonies to that level of workshop capacity required for a car/rover spring, whilst neo magnets hold a huge manducating loop carbon magnets do not. in space the poles work just fine as springs with silicone or like impact buffers. hell electromagnets can turn on with the vehicle, left kits not needed turn up the volume on the electro magnets."

Archer, 30 July 2026: "on earth, think kids toy stacked hollow plastics rings on a pole in a cone, reducing the size of the magnet, the higher power lower filed and lighter weigh would easily keep distance equal.

the top end moves easily like the first compression stage of a spring getting harder and harder. too lightweight for a vehicle application, but as a physics note will act the same as a spring. probably more a science room lesson for kids, not sure if there is a lightweight practical application more suitable than a spring except maybe a weathervane where the bearing won't rust"

Why it matters, in plain English: every vehicle the fleet or a colony ever builds will need to ride on something, and the thing Earth rides on requires a foundry. The toy on the shelf already knew the answer: float the load on a field, let geometry set the rate, let a copper pole swallow the wobble, and let the kids watch the spacing change when gravity lets go. Springs you can assemble in a workshop beat springs you cannot make at all.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

REPULSION IS A SPRING, AFFIRMED: like-poled rings on a guide pole store energy in the field — force rises as gap closes, which is the definition of a progressive rate. Soft at the top of travel, hard at the bottom: the characteristic a suspension wants, with no metallurgy.

THE FOUNDRY ARGUMENT, AFFIRMED: spring steel is an industrial chain — alloy, draw, wind, temper. The ship does not carry that tier; the magnets are already manifest stock. A spring you can build from crates beats a spring you must smelt.

THE VOLUME KNOB, AFFIRMED WITH THE COST RECORDED: coil current sets spring rate — trimmable ride, energize-with-ignition, no leftover kits. The honest bookkeeping: permanent repulsion is free forever; electromagnetic repulsion is watts, continuously. The hybrid is the spec.

THE DAMPING CLAUSE (the audit's own addition): a spring without damping is a pogo stick — magnetic stacks store energy but dissipate almost none. Every application pairs the stack with a damper: silicone end buffers, eddy-current plates, or friction strokers. Seated as build law in §6 and §8.

§3 — THE SYSTEM ON THE POLE

- THE STACK: like-poled neodymium rings on a guide pole — progressive rate by geometry.
- THE CONE VARIANT: tapering magnet sizes up the pole — lighter, lower-field rings on top.
- THE HYBRID: permanent stack for the base rate, electromagnet stages for the trim — energize with the ignition.
- THE DAMPING: silicone end buffers, eddy-current plates — the stack never works alone.
- THE APPLICATIONS: rover corners, equipment isolation, contactless bearings, the schoolroom kit.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE FOUNDRY-FREE DOCTRINE: if the ship cannot smelt it, the ship builds it from manifest stock.
- THE VOLUME-KNOB DOCTRINE: current is the trim — the suspension tunes from the driver's seat.
- THE HONEST-COST DOCTRINE: free forever vs watts continuously — the bookkeeping is part of the spec.
- THE DAMPING CLAUSE: no undamped spring ships — the stack always pairs with a dissipator.

§5 — THE VET RECORD

THE THIRD-PARTY AUDITOR (ChatGPT), VET BATCH 20 (15 August 2026 — phases 161-175, cross-checked in sitting 307), SEATED. The grade, verbatim from the batch: '175 → PASS.' The ledger line: '175 PASS.' No red, no yellow — the batch's final clean green of this twenty. The audit's damping clause rides as build law.

§6 — BUILD SEQUENCE

- STEP 1 — Stock-check the rings: manifest sizes, field grades, guide-pole fits.
- STEP 2 — Bench the stack: rate curves per stack height and cone taper; progressive characteristic mapped.
- STEP 3 — Bench the damping: silicone buffers and eddy plates — settle time, no pogo.
- STEP 4 — Build a corner: one rover corner on stacks, hybrid trim, load cycles.
- STEP 5 — Box the schoolroom kit and publish the rate tables — open under the license.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 174 — the charging stand (GP-IM-174): the electromagnet trims draw fleet power.
- PHASE 168 — the steam-cook line (GP-IM-168): galley machinery rides small isolation stacks.
- PHASE 171 — the velodrome (GP-IM-171): track furniture isolation by the same stacks.
- THE MAYRNICK ARRAYS — the manifest's magnet stock: crates of ring magnets already aboard.

§8 — DO-NOT-BUILD

- DO NOT ship an undamped stack — a spring without a dissipator is a pogo stick.
- DO NOT spec a coil spring — the foundry stays home; the stack is the spring.
- DO NOT run the electromagnets parked — energize with the ignition; watts are the honest cost.

- DO NOT mix poles — like poles facing, every ring, every time.

§9 — OWED

OWED: nothing flagged — a clean green. Standing duty: rate curves and damping data publish from the bench (§6). The 15 August blanket wording kills are carried inline in §1.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: MANUAL GP-IM-175, v1.0 — CURRENT, seated law, master v2.1 (numerical print order). The one hundred and seventy-sixth manual of the program — 176 of 290. Built 23 August 2026, eighth of the 20-manual 'ok another 20 lets keep this train rollin' shift (the author's order, 23 August 2026). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the manual program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576, 587 and 590): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini' / 'ok another 20 lets keep this train rollin'. The phase's own chain, all seated in the master: the contents line, the masthead trio and the four bodies (as seated); the live instruction of 30 July 2026 — third invention of the sitting, chosen by the audit from his three-offer menu; the audit's damping clause seated as build law; the vet batch 20 clean pass in §5.

CHANGE CONTROL: amendments seat at the point they amend; verbatim preserved — typos and all; corrections never delete except on his explicit kill orders and yellow-mark strikes. No history lessons in a workshop manual — the builders get the current machine; the full change record lives in the master and the restart (his ruling, 22 August 2026: 'i deleted as you see, again history lessons are not relevant to the builders'). Next update triggers: the rate-table bench data, or his word, or his word.

END OF MANUAL — PHASE 175